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Review

Fiber-Optic Gyroscopes in Modern Navigation Systems: A Comprehensive Review

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Abstract

This paper provides a comprehensive overview of the progress in fiber-optic gyroscope technology, covering 260 key studies of the last ten years. A critical comparative analysis of fiber-optic gyroscope with alternative inertial sensors (Micro-Electro-Mechanical Systems, Hemispherical Resonator Gyroscope, Ring Laser Gyroscope) has been carried out. Confirming the unique advantages of fiber-optic gyroscope for autonomous navigation. Fundamental limitations of accuracy are considered in detail: temperature drifts, polarization noise, and Rayleigh backscattering. Modern hardware methods for suppressing these errors, including the use of photonic crystal and hollow fibers (Air-Core/Hollow-Core), are also considered in this work. The central place in the review is occupied by the analysis of the technological paradigm shift from bulky discrete circuits to hybrid integrated photonics (Indium Phosphide, Silicon Nitride, Lithium Niobate) and hybrid architectures to reduce weight and size characteristics. The role of artificial intelligence (Deep Learning, Long Short-Term Memory) methods in nonlinear drift compensation and calibration is discussed. The usage of the Brillouin effect and optomechanics promising areas are outlined, necessary to create a new generation of navigation systems operating in the absence of Global Navigation Satellite Systems signals.

Keywords: fiber-optic gyroscope; navigation system; temperature drift; polarization noise; Rayleigh backscattering

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1. Introduction

The fiber-optic gyroscope (FOG) is a well-known orientation and angular velocity sensor [1,2]. Compared to other types of gyroscopes, the FOG has a recognized advantage in the form of low drift, the so-called angle random walk coefficient (ARW). ARW is an accuracy metric used to quantify the effect of high-frequency random noise on the gyroscope's output signal [1]. This is one of the most important parameters that determine the short-term stability and sensitivity noise of the sensor. Thus, the FOG ensures a low level of noise in measurements and helps to improve the accuracy of navigation systems.

The relevance of research in the field of FOGs is not limited to their wide application; there is always a need to further improve their accuracy and miniaturize their dimensions. Modern navigation systems require gyroscopes that can provide autonomy in the absence of GPS positioning, for example, in space, underground, underwater, or in heavily built-up urban environments. At the same time, the active development of Unmanned Aerial Vehicles (UAVs), robotics and machine learning create a demand for compact and inexpensive FOG [2] with high accuracy characteristics.

Despite the tremendous achievements, there are still some unresolved problems in the field of fiber-optic gyroscopy, such as the influence of temperature drifts [3–8], polarization noise [9–13] and Rayleigh backscattering [14–19]. These factors limit the actual parameters of the accuracy and reliability of the devices, which makes their further optimization the subject of active scientific research in this field.

The purpose of this review is a comprehensive analysis of modern FOGs and their role in modern navigation systems, as well as an analysis of methods for overcoming the effects of temperature drifts, polarization noise, and Rayleigh backscattering through the prism of new materials (hollow and stranded fibers) [20–22] and intelligent algorithms [23–25]. The principles of operation of various types of FOGs (interferometric and resonator) are considered, a comparative analysis of their advantages and disadvantages is carried out, and promising areas of development are identified, including error compensation methods and the use of new materials.

2. Materials and Methods

This paper systematizes and critically analyzes the progress in the field of FOG achieved by the world scientific community over the past decade. To ensure the representativeness of the sample, a literature search was performed in the IEEE Xplore, Scopus, and Web of Science databases with an emphasis on experimental research and highly rated publications (Q1/Q2).

Despite the fact that in recent decades, researchers have made significant efforts to solve FOG problems, and many complex problems have been solved, there is still a shortage of generalizing publications devoted specifically to error analysis. This article summarizes and analyzes the progress in recent work so that researchers in related fields can use the experience of their predecessors more effectively and contribute to the continuous improvement of instrument accuracy.

This review focuses on three fundamental limitations of accuracy, which remain the main challenges for developers: temperature drifts, polarization noise, and Rayleigh backscattering. The main sources of these errors include uneven distribution or temperature changes along the fiber (Shupe effect), imperfect polarization components and power leakage into undesirable polarization, as well as light scattering by microuniformities in the fiber. The suppression of these non-reciprocal optical errors is an important research topic, since their effective compensation guarantees stable output of navigation parameters in harsh operating conditions.

One of the important directions considered in the work is the analysis of the transition from classical hardware compensation methods to algorithmic ones, including the use of neural networks (Deep Learning, LSTM). Finally, studies of FOG schemes with high potential for practical application, aimed at reducing weight and dimensional characteristics (SWaP-C), are considered. In recent years, thanks to the development of integrated photonics (InP, SiN, $LiNbO_3$) and the emergence of new fiber types (PCF, Air-Core), several integrated gyro circuits have demonstrated attractive characteristics that are expected to form the basis for creating autonomous navigation systems on a chip.

This article is structured as follows:

Section 3 describes the evolution of gyroscopic technologies (MEMS, HRG, RLG) and provides a comparative analysis of advantages and disadvantages, determining their place in modern navigation systems. Section 4 provides a classification of FOGs into interferometric and resonator types with a description of the principles of their operation. Section 5 provides a detailed review of research on the three main sources of FOG errors, as well as an analysis of researchers' achievements in suppressing them. Section 6 discusses development prospects, including the transition to integrated photonics, the use of new materials and methods of digital signal processing (DSP and AI). In conclusion, the results are summarized and the key trends defining the future of inertial navigation are formulated.

3. The Evolution of Gyroscopic Technology

A gyroscope is a device capable of responding to changes in the angles of orientation of the body, where it is installed relative to the inertial reference system [26–28]. Gyroscopes are often used at many mobile stations to stabilize technical equipment, determine their navigation and dynamic motion parameters [29–31]. With data on the angular velocity and time of rotation, it is possible to calculate the angle of rotation. Knowing this data and having the calculated distances, you can plot the trajectory of the object, and then determine its current position and coordinates. Thus, it can be noted that gyroscopes are the second information component of navigation systems [32]. Having data on the initial coordinates and position of an object in space obtained from accelerometers and gyroscopes installed on this object, it is possible to calculate both its current position in space, as well as the direction of movement, speed and distance traveled [33]. Gyroscopes are divided into mechanical and optical ones according to the type of technology used for measuring. The main representatives of mechanical gyroscopes are Micro-Electro-Mechanical systems (MEMS), Hemispherical Resonator Gyroscopes (HRGs), and using optical technology—Ring Laser Gyroscopes (RLGs) and Fiber-Optic Gyroscopes (FOGs).

3.1. MEMS

MEMS are microscopic devices that can measure the angular velocity (rotational velocity) of an object. They combine tiny mechanical structures and electronics on a single silicon chip. These sensors can measure the speed of an object's rotation around a specific axis: 1-axis, 2-axis, and 3-axis (X, Y, and Z ordinates). MEMS gyroscopes use a vibrating, inert, micron-sized mass made using fine silicon processing technology. When rotation (angular velocity) occurs, the Coriolis effect causes secondary vibrations of this mass in a plane perpendicular to the primary vibrations. These secondary vibrations are detected by a capacitive sensor. Thus, the measured differential component is proportional to the angle of displacement. The resulting signal is amplified, demodulated and filtered, after which a voltage proportional to the angular velocity is applied to the output [34]. Upon closer examination of the basic principle of operation of these vibrating gyroscopes, each of them has a working body. When rotating, reciprocating motion is performed in one plane. If you place this body on a rotating platform, the plane of which coincides with the plane of oscillation, then the Coriolis force will act on the oscillating mass. Accordingly, the Coriolis force is directed perpendicular to the direction of oscillation and the axis of rotation [35].

In opposite directions of motion, the Coriolis force also acts in opposite directions [35]. This is the basis of the principle of a vibrating gyroscope [36]. Having determined the Coriolis force and knowing the linear velocity of the body, it is easy to calculate the angular velocity and its change, that is, the angular acceleration. According to Figure 1, Ω_Z is the angular velocity vector, which shows the axis of rotation (vertically upward) and the direction of rotation of the disk; V is the vector of the linear velocity of an object on the disk that is trying to move directly from the center to the edge (radially); F_c is the force of inertia acting on the object.

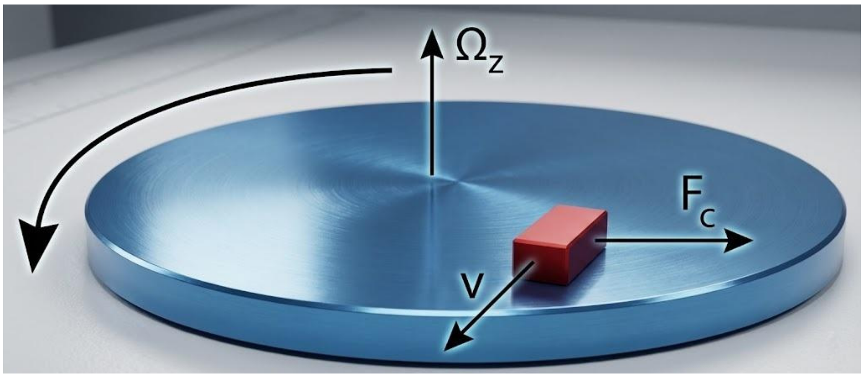

Figure 1. The principle of operation of a vibrating gyroscope.

All designs of vibrating gyroscopes, with their wide variety, can be reduced to several types. Beam gyroscopes were one of the very first. Their principle of operation is as follows: the cantilever beam (plate) is made to oscillate with the help of piezoelectric elements in the direction of the X axis (Figure 2).

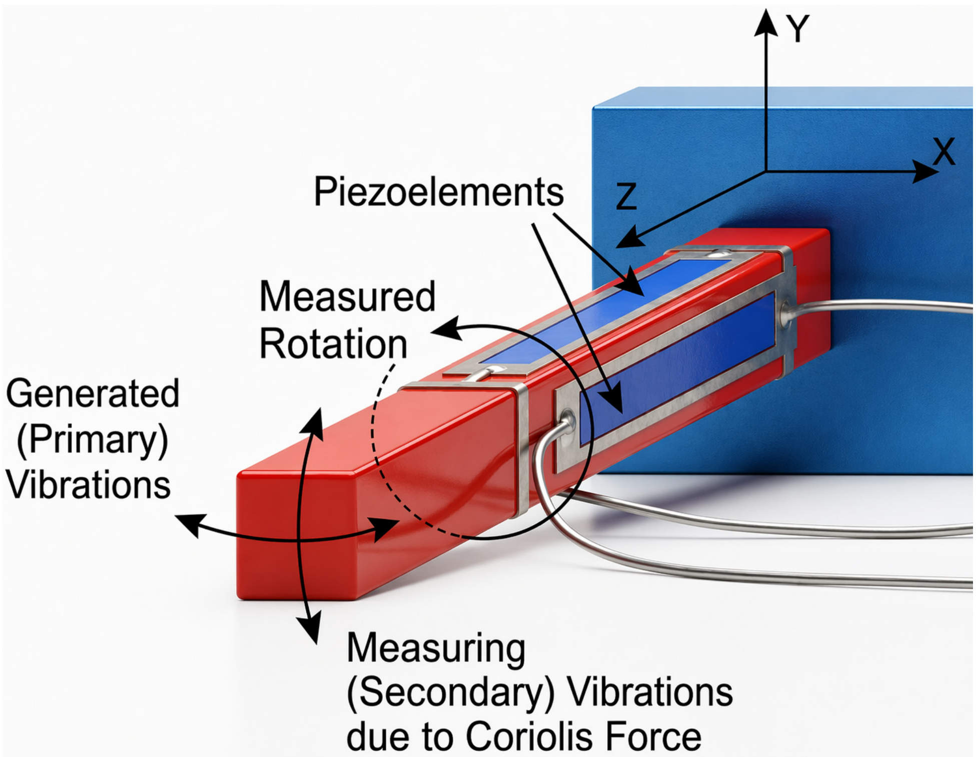

Figure 2. The principle of operation of a beam gyroscope.

Under the action of the Coriolis force, when rotating relative to the Z axis parallel to the longitudinal axis of the beam, vibrations along the Y axis are excited. Although beam gyroscopes were originally used for expensive military purposes, they are now also being used in low-cost commercial consumer electronic applications for the automotive, defense, industrial, and medical industries [37].

The main disadvantage of MEMS gyroscopes is their limited accuracy and stability. Even in the absence of angular velocity, the gyroscope can produce a non-zero signal (offset).

This displacement is unstable and drifts with time and under the influence of external factors the so-called offset drift or zero offset (Figure 3). This is one of the main sources of error that accumulates when integrating angular velocity to determine orientation. The temperature change strongly affects the zero signal and the scale factor of the gyroscope, causing temperature drift. To compensate that, complex algorithms and calibration are required. MEMS gyroscopes can give erroneous readings when exposed to strong linear accelerations (shocks, vibrations), which is critical in high-dynamic applications.

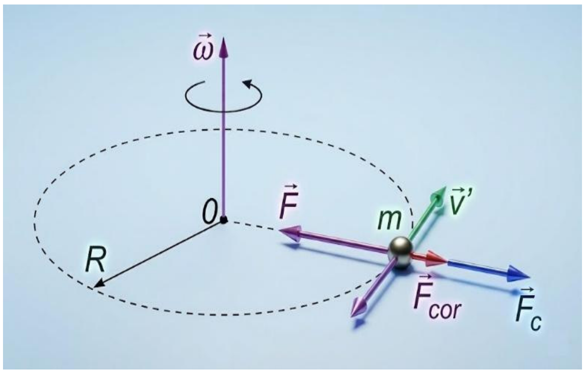

Figure 3. The dynamics of an object in a non-inertial system.

That is, additional mechanical vibrations can cause additional noise in the measured signal, reducing accuracy and resolution. Despite these disadvantages due to their ultra-compact size, low cost, and low power consumption, MEMS gyroscopes have found wide application in consumer electronics (smartphones, tablets), the automotive industry, small aircraft (drones, UAVs) [38–40], and many other fields. In these industries, accuracy is considered sufficient, but their drift and vibration sensitivity limit their use in strategic navigation, despite progress in calibration [34,41,42]. The increased demand for mobile devices is also the reason for the growth of the MEMS gyros market (Figure 4). The cost of MEMS gyroscopes is expected to decrease in the coming years, leading to increased use of these devices.

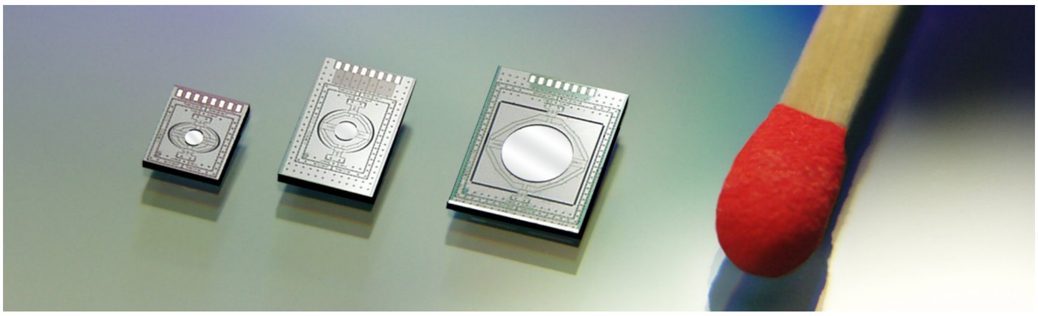

Figure 4. Modern MEMS gyroscopes.

3.2. HRG

HRG (Figure 5), first introduced by Delco in 1982, is a compact solid-state sensor that occupies a key niche between optical and MEMS gyroscopes [43]. The principle of its operation is based on the inertial properties of standing waves excited by electrostatic forces at the edge of a thin-walled resonator made of fused quartz. The device has no rubbing parts, which provides it with unique impact resistance, vibration resistance and insensitivity to radiation [34,41,42]. Due to its exceptional reliability (a service life of up to 15 years with a 99.5% uptime probability) and high accuracy, HRG has become the main component of inertial systems for satellites, spacecraft, strategic missiles and naval vessels, displacing many mechanical analogs [26,43,44]. Despite the complexity of manufacturing precision quartz spheres, which limits the range of manufacturers (global brands—Northrop Grumman, Safran, Raytheon), the introduction of planar electrodes reduces the cost of production for mass applications [45–47]. In parallel with the evolution of solid-state gyroscopes, in the second half of the 20th century, the invention of optical fiber gave a powerful impetus to the development of photonic measurement technologies, opening a new era in navigation.

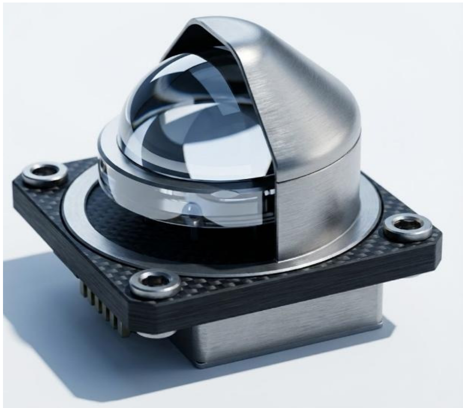

Figure 5. Hemispherical Resonator Gyroscope.

3.3. RLG

Ring Laser Gyroscopes (Figure 6) based on He-Ne lasers appeared in the 60s of the last century as the first type of optical gyroscopes, the operation of which, as is known, is based on the Sagnac effect. Thus, using RLGs, it is possible to directly obtain the value of the angular displacement, while the traditional model of the later FOG primarily measures the value of the angular velocity based on the phase shift in the oncoming light waves. Another difference from FOG is that the RLG has an oscillator in the form of a so-called "frequency stand", which allows it to increase its sensitivity and, consequently, the accuracy of measurements. The frequency stand is usually located in the center of the RLG (resonator) housing with optical channels that act as light guides. Monoblock resonators are made of materials with a low temperature coefficient of linear expansion: invar, fused quartz, citall and constasyl. The light guide is made in the form of a gas-discharge tube filled with a mixture of isotopes ^{3}He and ^{20}Ne, ^{22}Ne. The value of λ = 0.63 microns (630 nm) is recognized as the most optimal operating wavelength of lasers in RLGs. RLGs have excellent linearity, but require mechanical dithering to eliminate frequency capture, which reduces service life [48,49].

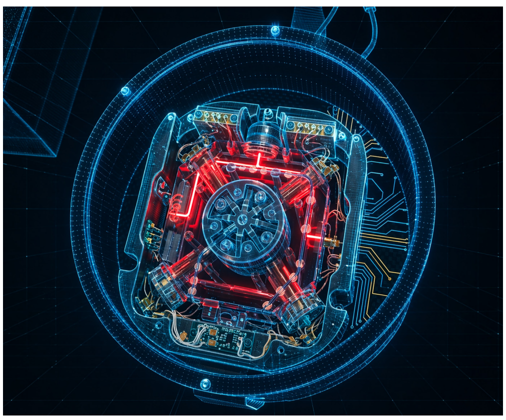

Figure 6. Ring Laser Gyroscopes.

3.4. FOG

Gyroscopes are widely used in stabilization and navigation systems in various industries: aviation, shipbuilding, production of UAVs, robotics, railway transport, oil and mining industries, defense industry, rocket and space engineering, automotive, medicine, etc. FOG is an angular velocity sensor that uses the Sagnac effect to measure the rotational speed of an object, and has established itself as a reference angular velocity sensor for applications requiring high reliability and long-term stability [48,50–54]. This effect consists of the fact that the light passing through the fiber ring in two opposite directions acquires a phase difference proportional to the angular velocity of rotation of the ring. This phase difference, or the Sagnac shift, is measured as a shift in the interference pattern. According to this principle (Figure 7), when two light waves collide in an optical fiber coil during its rotation, one wave will acquire a Sagnac phase shift of $+\varphi s/2$, and the other $-\varphi s/2$. The dependence of $\Delta\varphi s$ on the angular velocity of rotation of the coil Ω is expressed by the Formula (1):

$$\Delta\varphi s = \frac{2\pi LD}{\lambda c}\Omega, \tag{1}$$

where λ is the light wavelength (weighted average); c is the propagation velocity of optical radiation in vacuum; L is the length of the fiber in the coil; and D is the diameter of the coil. The product LD is usually called the effective area of the Sagnac interferometer. The larger it is (due to the length of the fiber or the diameter of the coil), the higher the sensitivity of FOG [55–57].

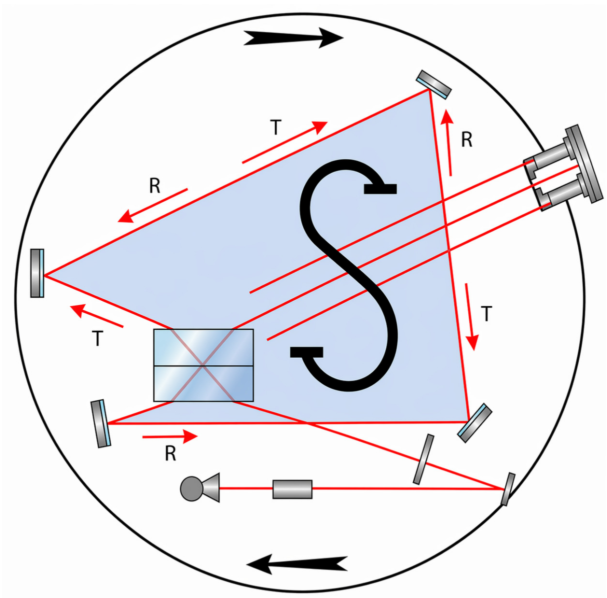

Figure 7. The scheme of the experience of a Sagnac.

The phase difference $\Delta\varphi s$ of the oncoming waves is determined by the intensity of the light I at the photodetector, which will vary according to Formula (2)

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\Delta\varphi s, \tag{2}$$

where I_1, I_2 are the intensities of the oncoming waves. The cosine dependence of I on $\Delta\varphi s$ leads to a low sensitivity of I to low angular velocities; therefore, to increase it, a special phase modulation by $\pi/2$ is introduced, which translates the output signal of the FOG to the point of maximum sensitivity at zero angular velocity [57].

The ultimate sensitivity of FOG is limited by the photonic, electronic, and shotgun noise of the radiation source and receiver. It can be seen from Formula (1) that by changing the length of the fiber, it is possible to obtain a gyroscope with different sensitivity (resolution) when different levels of the effective angular velocity correspond to the same measured phase difference. However, the range of measured speeds is limited, but this problem is solved by using feedback output signal processing circuits and special algorithmic methods [58].

The physical essence of this effect lies in the fact that when the annular fiber-optic resonator rotates, there is a slight difference in the optical path traversed by clockwise (CW) and counterclockwise (CCW) light waves propagating in the contour. In addition, the optical travel difference has a linear relationship with the rotational speed at the input, so it can be used to determine the rotational speed of the carrier. Compared with mechanical and laser analogs, FOG has a fundamental advantage in the form of an ultra-low coefficient of random walk in an angle (ARW) and the absence of dead zones (lock-in) [59–62]. These characteristics are critically important for inertial navigation systems (INS) operating in the absence of GNSS signals: in deep space, underwater, during underground construction and in electronic warfare [63–65]. However, miniaturization creates new challenges that require innovative approaches to architecture, which is confirmed by the creation of prototypes of navigation systems based on commercial components [66] and new compact models [67]. The development of UAVs, nanosatellites, and robotics creates demand for sensors that combine the precision of FOG with the dimensions of MEMS [47,57,68–70].

3.5. New Types of Fibers

The classic single-mode fibers are being replaced by specialized ones. Photonic crystal fibers (PCF) (Figure 8a) [71] and Air-Core fibers (Figure 8b) radically reduce thermal and radiation sensitivity, allowing us to create integrated solutions with high stability [72], as well as effectively solve the problems of losses in nodes communications [73]. The use of PCF resonant circuits (RPCFOG) opens the way to miniature sensors with a drift of less than 0.5°/h [74].

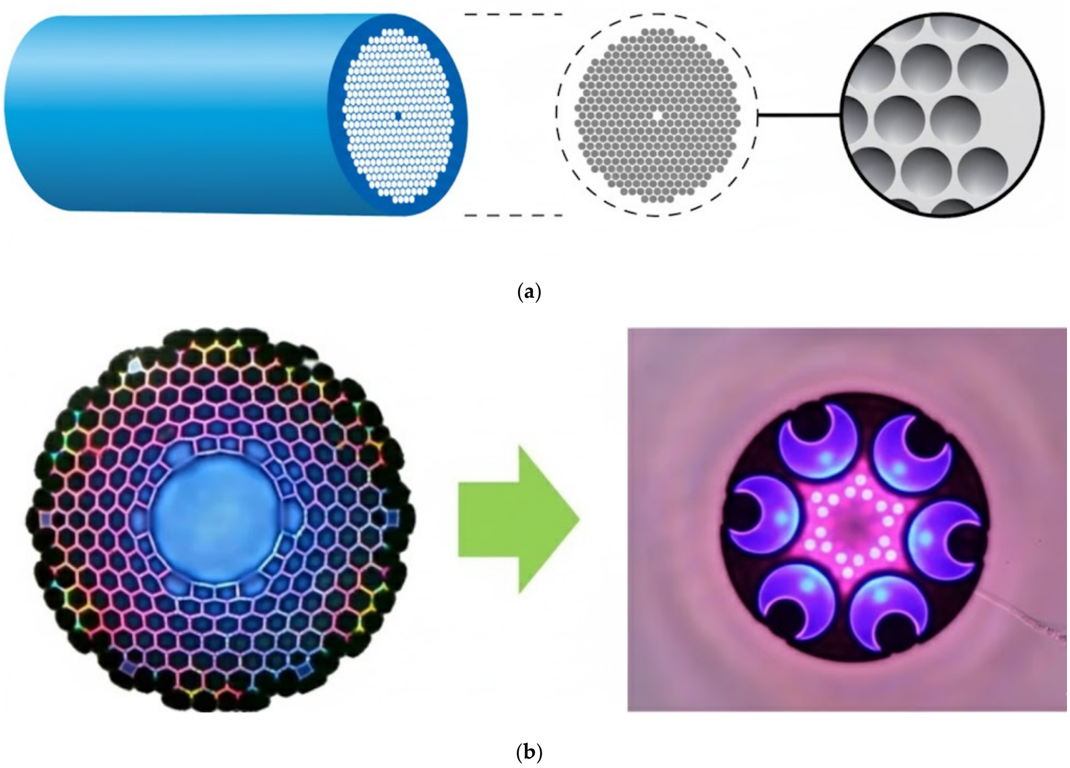

Figure 8. New types of fibers: (**a**) Photonic crystal fiber; (**b**) Air-Core fiber.

4. Classification of FOGs

According to the principle of useful signal extraction, there are two main types of FOG—interferometric (IFOG) and resonator (RFOG). The principle of operation of an IFOG is based on direct measurement of the phase shift due to the interference of oncoming waves. The most advanced technology using long coils (up to 20 km) to achieve record sensitivity [22,75–77], IFOGs are indispensable for applications in fiber-optic sensors, as they have enormous advantages, such as full solid state, low cost, wide dynamic range and high sensitivity to environmental factors. It is widely used in various fields such as inertial navigation, seismic monitoring and precision measurements. Due to the continuous improvement of optical components in the field of FOG research, the industry is placing ever higher performance requirements on IFOG. Optimization of sensitivity and stability remains key among the various directions of IFOGs development. Considering the size and cost, noise reduction has become the optimal choice for improving IFOG performance.

RFOG is an inertial sensor with great application potential, with higher advantages in optical stability and miniaturization of the system. It uses short, high-quotient resonators, promising miniaturization to the chip level, but requiring sophisticated coherent noise suppression techniques [18,78–83]. Like other optical gyroscopes, the RFOG is also based on the Sagnac effect to measure the rotational velocity of the carrier. Unlike the IFOG, the RFOG can significantly shorten the length of the sensitive fiber-optic contour due to the multipath interference effect of the fiber-optic resonator. This is essential for miniaturizing the system and increasing its ability to interfere with the interference environment.

To clearly delineate the operational trade-offs between the two primary FOG architectures, Table 1 provides a direct comparison between IFOG and RFOG.

Table 1. Comparative analysis of IFOG and RFOG architectures.

Characteristic	IFOG	RFOG
Operating Principle	Phase difference measurement	Resonant frequency shift measurement
Required Fiber Length (Size)	Large (100 m to several kilometers)	Miniature (1 m to 20 m in a ring resonator)
Sensitivity	Extremely High (Navigation and Strategic grade)	Moderate to High (Tactical grade)
Scattering Noise Sensitivity	Moderate (Effectively suppressed by broadband SLDs)	Extremely High (Requires highly coherent lasers, leading to severe coherent backscattering issues)
Typical Applications	Submarines, Spacecraft, Strategic Missiles	Miniaturized UAVs, Autonomous vehicles, Tactical systems

As summarized above, while IFOGs remain the undisputed standard for ultra-high-precision navigation due to their mature noise suppression techniques, their reliance on kilometer-long fiber coils fundamentally limits their miniaturization. Conversely, RFOGs offer a pathway to chip-scale integration by utilizing short resonant loops. However, the requirement for highly coherent lasers in RFOGs dramatically exacerbates Rayleigh backscattering and Kerr-effect noise, remaining the primary bottleneck preventing RFOGs from reaching navigation-grade sensitivity.

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5. Problems and Research Gaps

Temperature drift, polarization noise, and Rayleigh backscattering are the three fundamental factors limiting the accuracy and long-term stability of FOGs. This section provides a detailed analysis of the mechanisms of influence of these effects on the metrological characteristics of the device.

5.1. Influence of Temperature Drifts

Shupe effect, which is the main source of temperature drift in FOG caused by temperature gradients along the fiber, remains the main source of errors in FOG [6,84,85]. It occurs due to a temporary violation of the reciprocity of the path of oncoming light rays in the fiber coil when the temperature changes. Since a fiber coil consists of many small fiber cores, in ideal conditions, the oncoming rays travel through each such section in the same way. However, with rapid temperature changes (heating or cooling), different sections of the fiber can heat up/cool down at different rates. This phenomenon's mechanism is a local change in the optical path length, through to the thermo-optical effect and thermal expansion of the fiber. However, due to the finite velocity of radiation propagation, the oncoming waves pass through the perturbed area at different points in time, experiencing unequal phase delays. The resulting non-reciprocal phase shift, misinterpreted by the system as a rotation signal, correlates not with the absolute value of temperature, but with the rate of its change over time, which determines the drift of the zero signal and limits the maximum accuracy of the device under dynamic conditions.

Figure 9 shows a fiber coil, part of which (red) is heated and part (blue) is not, creating a temperature gradient. A CW and CCW beam pass through these zones. Due to uneven heating, the refractive index (n) and length (L) vary differently for oncoming rays. As a result, a non-reciprocal phase shift ($\Delta\varphi_{Shupe}$) occurs, which is detected as a false rotation signal (ERROR).

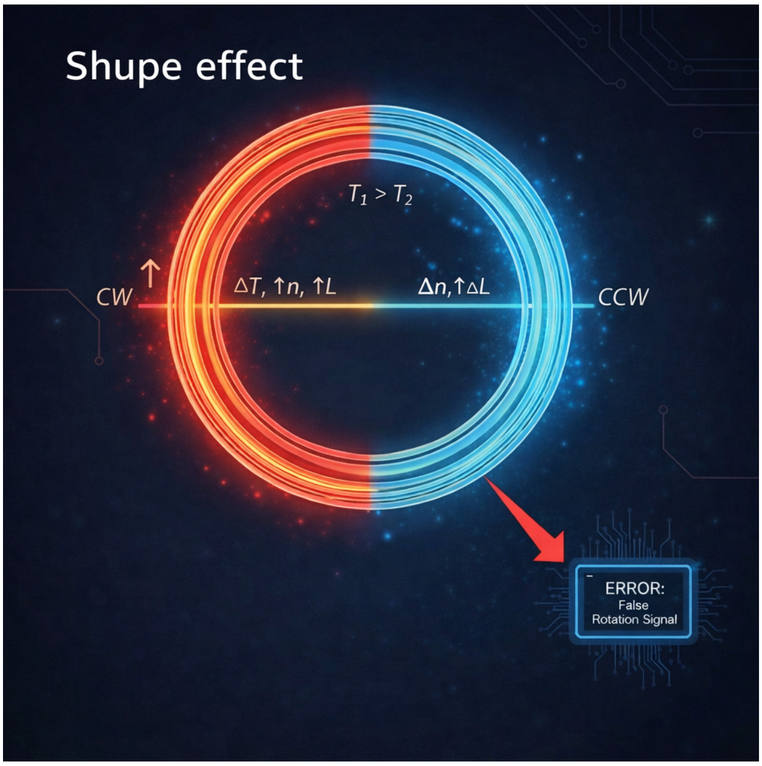

Figure 9. Shupe effect.

Multiphysical models (CFD/FEM) have been developed for the analysis of thermal fields in vacuum [64] and complex coil designs [86,87]. The thermo-optical properties of sensor materials are widely studied in real time [88,89]. By optimizing the heat sink, it becomes possible to reduce the temperature of the components by 78% [90]. 3D models for vacuum [64], large coil analysis [86], and winding tomography [87] are used. Humidity has a critical effect, causing mechanical drift of the fiber [91,92], which also requires taking into account the effect of humidity on the frequency response of modulators [93]. Optimization of the heat sink reduces the heating of components [90], and integrated sensors on hollow fiber [94], platforms for silicon chips [88], as well as SPR methods [89] are used for temperature control. Methods and technologies for minimizing temperature drift of FOGs are shown in Table 2.

Table 2. Methods and technologies for minimizing temperature drift of FOGs.

Category	Specific Technique	Mechanism	Effectiveness and Pros	Refs.
Winding Techniques	Double Quadrupole	The symmetrical distribution of fiber coils ensures mutual compensation of phase shifts from radial temperature gradients	Reduces temperature drift up to 13 times compared to conventional winding. It is an industry standard for navigation systems	[84,95,96]
Advanced Fibers	Hollow-Core/Air-Core	Light propagates in air/vacuum, where the thermo-optical coefficient (dn/dT) is orders of magnitude lower than in quartz glass	Provides minimal sensitivity to temperature and radiation. Radically reduces the “thermal inertia” of the coil	[20,21,73,97–100]
	Thin Diameter Fibers, 60–80 µm	Reducing the fiber diameter reduces the volume of the epoxy resin and improves the thermal conductivity inside the coil	Reduces temperature gradients due to faster temperature equalization over the volume of the coil	[72,86,101]
Thermal Control	Heat Dissipation Structure	The use of CFD modeling for the design of radiators and heat-conducting enclosures	It reduces the temperature of critical components by 78%, preventing local overheating	[90]
Algorithmic/AI	Deep Learning: LSTM, RNN	Training neural networks on time series to predict nonlinear sensor hysteresis, which is difficult to describe in formulas	It surpasses traditional polynomial models. Effectively compensates for drift in conditions of rapid temperature changes	[25,102]
	SVR + VMD	The combination of the support vector regression (SVR) with variational mode decomposition (VMD) for noise filtering	Allows you to separate the useful signal and the temperature trend, improving the stability of the displacement	[8,23,86]

5.1.1. Hardware Compensation

The use of stranded (MCF) [22,103] and thin (60 microns) fibers [101,104,105], as well as fibers with an elliptical core (Panda) [106] reduces sensitivity to gradients. The breakthrough was the use of hollow fibers (Air-Core) having a minimum thermo-optical coefficient [21,97,107]. The "double quadrupole" winding scheme reduces the drift by 13 times [84]. Annealing of $LiNbO_3$ crystals increases the stability of MIOC chips [108,109].

5.1.2. AI-Compensation

Neural networks (LSTM, RNN) and hybrid models (SVR + VMD, MPE-CEEMDAN) effectively simulate hysteresis and nonlinearities, surpassing polynomial methods [7,8,23,25,102,110–116]. To quantitatively substantiate the paradigm shift towards deep learning for thermal drift compensation, it is necessary to compare the performance of complex models like LSTM and RNN with traditional compensation techniques. Historically, Polynomial Fitting (PF) has been the industry standard. However, PF models assume a memory-less, static mapping between temperature and phase drift. This approach inherently fails to capture the complex thermal hysteresis present in FOG coils, where the drift depends not only on the absolute temperature but also on the thermal history and the rate of temperature change (dT/dt). In contrast, sequence-based deep learning models, particularly LSTMs, utilize internal hidden states to retain temporal dependencies. This allows them to effectively model the dynamic hysteresis loop. Table 3 summarizes the comparative performance data of PF and LSTM/RNN models aggregated from recent benchmark studies.

Table 3. Performance comparison of thermal drift compensation methods.

Compensation Method	Memory/Hysteresis Modeling	Typical RMSE Reduction	Typical Bias Stability (Dynamic Temp)
Uncompensated	None	0%	~0.02°/h
Polynomial Fitting	No (Static mapping)	40–50%	~0.01°/h
LSTM/RNN Architectures	Yes (Temporal dependencies)	80–95%	<0.005°/h

While optimized traditional PF typically reduces the Root Mean Square Error (RMSE) of temperature drift by up to 50%, LSTM-based architectures consistently achieve an 80% to 95% reduction. Furthermore, under highly dynamic temperature gradients, deep learning compensation frequently improves the bias stability to better than 0.005°/h, significantly outperforming PF, which typically plateaus around the 0.01°/h mark due to its inability to dynamically adapt to the hysteresis effect [25,102].

Methods of active modulation control for thermal compensation have been proposed for dual-channel systems and RFOG [117,118]. The method of optimizing the modulation parameters using a particle swarm (PSO) also reduces noise [119]. Stochastic models of degradation along three axes [120] and optical path measurement (ToF) methods for correcting the scale factor [121] have been developed.

5.2. The Occurrence of Polarizing Noise

Polarization noise occurs due to unwanted switching of the light signal between different states of polarization inside the optical fiber. FOG uses polarization-preserving fibers (PM fibers), or polarizers, to ensure that light propagates in only one strictly defined state of polarization. Zero-offset drift is frequently caused by parasitic interference signals resulting from the cross-coupling of polarization modes. This phenomenon is typically triggered by external factors like temperature shifts and vibrations, as well as physical inconsistencies such as splice faults or microscopic fiber defects [11,12,95].

Figure 10 shows a fiber coil where the main signal propagates. However, due to various factors, some of the light switches to a parasitic state of polarization. The stray light can return to the ground state by the interfering. The results are polarization drift and noise.

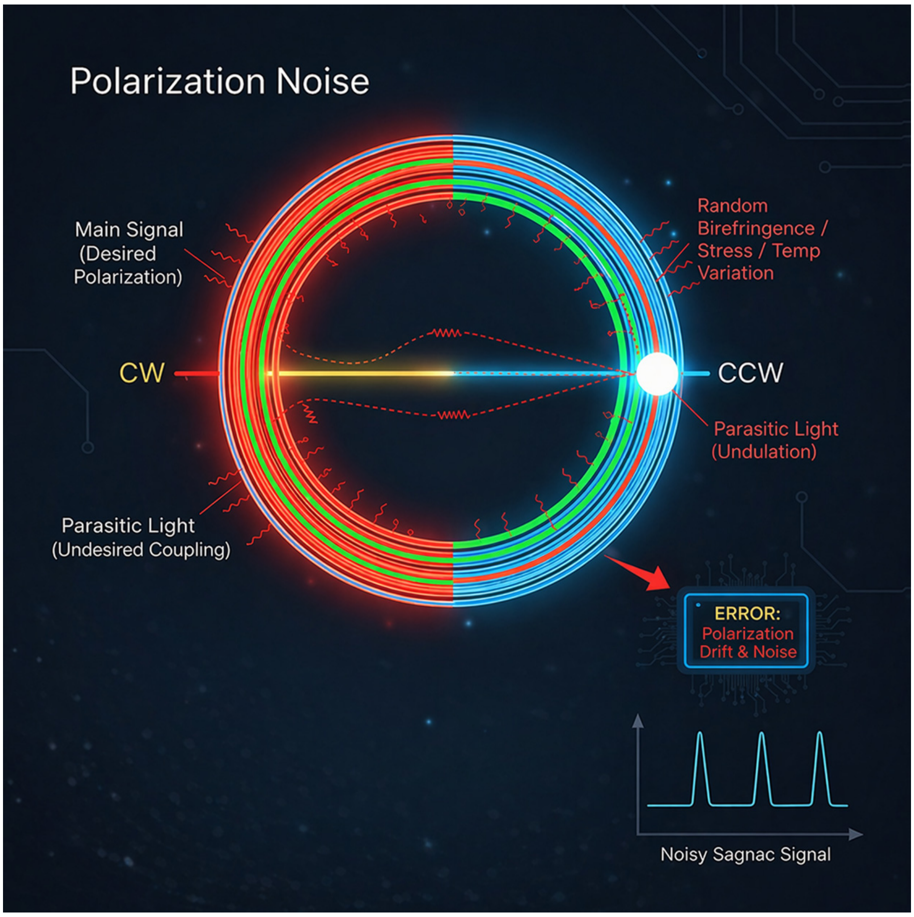

Figure 10. Polarization noise.

Phase birefringence within the optical fiber induces a differential phase shift in the orthogonal mode. The subsequent cross-coupling of this radiation back in the main mode generates coherent interference, which superimposes a parasitic signal onto the Sagnac phase. This manifests as both zero-bias drift and stochastic noise (green color in circle). Crucially, the severity of this error correlates with the optical source's coherence length: while high-coherence lasers increase parasitic mode interference, the deployment of broadband sources, such as Superluminescent Diodes (SLDs), effectively decorrelates phase errors and suppresses polarization-induced bias. To further decrease environmental sensitivities, including magnetic errors and the Kerr effect, techniques involving spun fibers and 90° axis-rotation schemes are widely adopted [11,13,122–124]. Inline polarizers and specialized couplers based on Photonic Crystal Fibers (PCF) ensure high extinction ratios [98,125–127]. Recent developments include hybrid single polarization resonators achieving stabilities of 0.7°/h [128], and Polarimetric FOGs (PFOGs) that analyze Stokes parameters to bypass the need for phase modulation [129]. The integration of active self-compensation via liquid crystal modulators and AI-driven algorithms is currently pushing accuracy toward the theoretical Sagnac limit [130], while in Whispering Gallery Mode (WGM) microresonators, coupling angle optimization has successfully reduced drift to 0.5°/h [131,132]. Methods and technologies for minimizing polarization noise of FOGs are shown in Table 4.

Table 4. Methods and technologies for minimizing polarization noise of FOGs.

Category	Specific Technique	Mechanism	Effectiveness and Pros	Refs.
Passive/Fiber	PM Fibers	The use of highly birefringent fibers (such as Panda or Bow-tie) for rigid fixation of the polarization state	They provide a basic level of protection against cross-coupling modes, but are expensive and sensitive to tension during winding	[106,124,125]
	Spun Fibers	Introduction of rotation of the axes of fiber anisotropy during the drawing process	Effectively average birefringence, reducing magnetic sensitivity and polarization errors	[21,95,133]
Assembly Techniques	90° Rotated Splicing	Welding of fiber segments with 90-degree rotation of the anisotropy axes to compensate for the accumulated phase difference	Compensates for the temperature dependence of birefringence and suppresses parasitic interference	[11,13,134]
Components	In-line Polarizers	Installation of polarizers with a high extinction coefficient directly into the optical circuit	The "parasitic" light is filtered, preventing its interference with the main signal. They are critically important for RFOGs.	[11,13,126,135]
	Specialized Couplers	The use of unipolarization couplers based on photonic crystal fibers (PCF)	They provide high mode selectivity and reduce communication losses	[74,136–138]
Advanced Architectures	P-FOG	Measuring the Stokes parameters of a light wave instead of directly measuring the phase	Eliminates the need for complex phase modulation by simplifying the optical circuit	[129]
Active/AI	Active Self-Compensation	The use of liquid crystal modulators and real-time control algorithms	It allows you to adjust the polarization, bringing the accuracy of the device closer to the theoretical limit of Sagnac	[130]

5.3. Rayleigh Backscattering and Source Noise (RIN)

Rayleigh backscattering is a phenomenon in which light propagating in a fiber is scattered backwards due to microscopic irregularities in its structure. Optical fiber, although it appears to be homogeneous, actually has random, very small fluctuations in the density of the material (irregularities smaller than the wavelength of light). When light passes through such inhomogeneities, a small part of it is scattered in all directions, including back towards the source. Coherent noise (RIN, backscattering, laser phase noise) limits sensitivity and causes frequency capture in RFOG [19,139,140].

In Figure 11, each of the counter rays (clockwise and counterclockwise) generates its own set of backscattered waves. The problem occurs when the scattered light from one beam (for example, from a clockwise beam) interferes with the main, non-scattered light of another beam (counterclockwise), which goes in the same direction as the scattered light. This interference creates random, fluctuating phase shifts that are added to the useful Sagnac signal. This manifests itself as the noise of ARW and worsens the stability of zero.

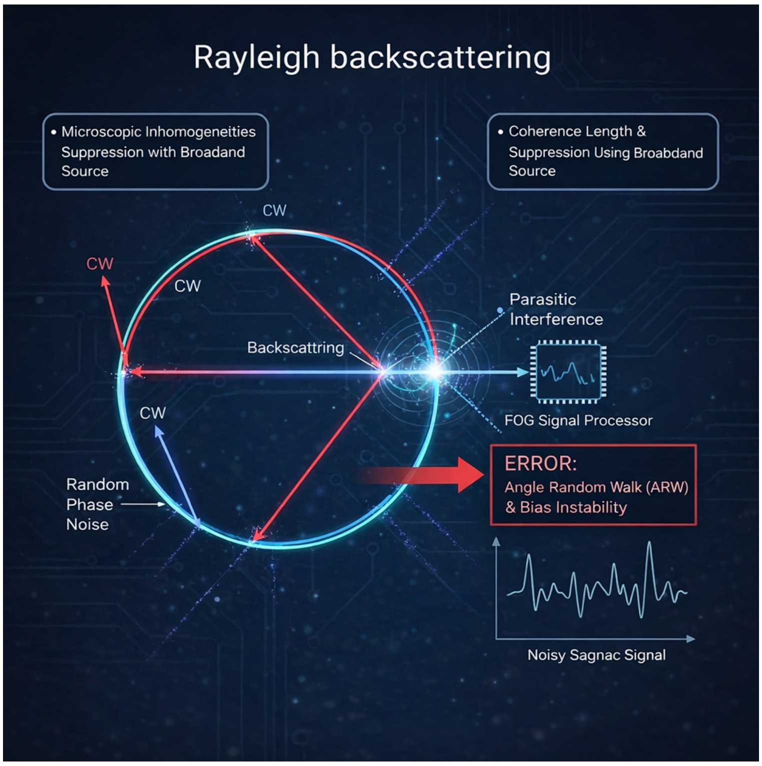

Figure 11. Rayleigh backscattering.

The higher the coherence of the light source, the more efficiently scattered waves can interfere with the main signal, amplifying this noise. The use of broadband light sources with a short coherence length is the main way to suppress this effect, as it reduces the likelihood of coherent interference.

While the use of broadband light sources such as SLDs is standard practice in modern FOGs, the underlying physical mechanism of noise suppression warrants a rigorous quantitative explanation. The fundamental issue arises from the interference between the primary signal wave and the parasitic counter-propagating wave generated by Rayleigh backscattering [141].

This interference only induces a non-reciprocal phase error if the backscattered wave is coherent with the primary wave. For a light source with a specific coherence length (L_c), coherent addition only occurs if the scattering event takes place within a strictly defined coherent scattering zone. Mathematically, for a fiber coil of total length L, the optical path difference between the interfering waves is less than L_c only for scattering centers located at a distance z from the coil's midpoint satisfying the condition $|L - 2z| < L_c$. Consequently, the coherent scattering zone is spatially strictly localized to the exact midpoint of the fiber loop, and its spatial extent is exactly equal to L_c.

Scattering events outside this microscopic region add incoherently, contributing to background intensity noise rather than phase fluctuations. Therefore, the RMS of the non-reciprocal phase error, $\Delta\phi_E$, is proportional to the square root of the ratio of the coherence length to the total fiber length (3):

$$\Delta\phi_E \propto \sqrt{\frac{L_c}{L}} \tag{3}$$

By employing an SLD with an ultra-short coherence length ($L_c \approx 30\ldots50$ μm) in a typical FOG coil (L ≈ 1000 m), the effective coherent scattering region is reduced by several orders of magnitude compared to narrow-linewidth lasers. This extreme spatial localization shrinks the ratio L_c/L to approximately 10^{-7}, systematically suppressing the coherent Rayleigh noise to navigation-grade levels.

Methods and technologies for minimizing the Rayleigh backscattering of FOGs are shown in Table 5.

Table 5. Methods and technologies for minimizing Rayleigh backscattering of FOGs.

Category	Specific Technique	Mechanism	Effectiveness and Pros	Refs.
Light Source Control	Broadband Sources: SLD	The use of sources with a short coherence length reduces the likelihood of interference of scattered waves with the main signal	The main method of coherent noise suppression in IFOG	[141]
	Laser Frequency Stabilization	Using feedback loops to stabilize the laser frequency and reduce phase noise	Compensates for phase noise and intermodulation distortion in high-precision circuits	[142]
Architecture Design	Fusion Schemes	Combining signals to subtract common-mode source noise (RIN) and external disturbances	Noise reduction by 20–27 dB, improved signal-to-noise ratio	[143–147]
	Composite Structures	The use of delay lines and polarizing multiplexing	Effectively suppresses RIN by separating noise and useful signal in time or polarization	[61,148,149]
Modulation and Detection	Period Overlapping	A special modulation in which the scattered waves are averaged over the detection period	Critically important for RFOG, it eliminates the coherence of scattered light	[79,82,150,151]
	Three-frequency Detection	Using sideband modulation to capture resonance and detuning from scattering frequencies	Prevents frequency capture (lock-in effect) and increases linearity in resonator gyroscopes	[152]
	PSK	Using phase modulation to decouple fluctuations in intensity and useful signal	Effectively separates the useful rotation signal and spurious amplitude noise	[99]
Filtering and Processing	Self-adaptive Filtering	The use of optical and digital filters that adjust to the noise parameters in real time	Minimizes the impact of source degradation and environmental conditions on accuracy	[109,120,138]

Suppression of RIN

Composite structures with delay, polarization [153,154] and circuits with a common Fusion source [143–147] gave a chance to reduce the noise by 20–27 dB. Self-adaptive modulation [155] and optical filters [61,148,149], are minimize the influence of the source. SLD power management eliminates drift from current fluctuations [141]. Simultaneous suppression of thermal noise and RIN makes it possible to achieve ARW of 15 $\mu grad/\sqrt{h}$ [76,77].

5.4. RFOG and Scattering

The main mechanisms of noise in RFOG are systematized in detail in the reviews [156]. Three-frequency detection [157], reciprocal modulation [158–160], sideband blocking [139], and high-frequency PDH modulation [161] are used in resonant circuits. Methods of suppressing external reflections from the ends of the fiber [162,163] and the frequency capture effect [120] are critical for accuracy. The PSK method is effectively used to decouple fluctuations in intensity and modulation [99]. Optimization of resonator scanning parameters also reduces errors [164]. The method of "superimposing periods" with broadband sources eliminates the coherence of scattered waves [79,82,150,151]. Active laser frequency stabilization and the use of multi-laser circuits compensate for phase noise [100,165–170]. The analysis of laser noise shows their effect on intermodulation distortion [142]. The Kerr effect is suppressed by modulation of 50% of the borehole [171].

6. Development Prospects

6.1. Integrated Photonics and Miniaturization

The downward trend in SWaP-C is driving the transition to integrated optical circuits (IOC) and hybrid assemblies. Multifunctional integrated circuit chips (MIOC) based on $LiNbO_3$ and InP combine sources, detectors and modulators, achieving stability of 0.018°/h on an area of less than 50 mm^2 [136,172–176] as well as tactical modules with record integration [177]. Specialized photonic integrated circuits have been developed for tactical applications [137]. Phase modulators with Faraday rotators eliminate frequency dependence [178]. Circuits with ultrashort coils (5 m) based on non-reciprocal modulators have been developed [179].

6.2. Silicon Photonics and SiN

The silicon photonics (Si-photonics) and silicon nitride (SiN) platforms create the technological base for the development of cost-effective and small-sized gyroscopes. The usage of the CMA-ES algorithm to optimize the geometry of spiral waveguides from SiN made it possible to achieve a Q-factor of Q > 60 million, which ensured tactical-class accuracy in passive operation [12,134,180–185]. A further reduction in optical losses is associated with the prospect of using bound states in a continuum (BIC) in the structures of spiral waveguides [186,187].

6.3. New Architectures

Coupled resonator cascade gyroscopes (CROW) [188] and anti-PT symmetric circuits [189] offer new ways to scale sensitivity. The implementation of anti-PT symmetric circuits represents a paradigm shift in optical gyroscope design due to their ability to fundamentally alter the mathematical scaling of the Sagnac effect. In traditional FOGs and standard resonant gyroscopes, the Sagnac-induced frequency splitting ($\Delta\omega$) is strictly linear with respect to the rotation rate (Ω), meaning $\Delta\omega \propto \Omega$. Anti-PT symmetric systems consist of coupled optical resonators with balanced gain and loss (or specifically tailored conservative couplings) that can be tuned to operate exactly at an Exceptional Point (EP)—a singularity in the parameter space where both the eigenvalues and eigenvectors of the system coalesce. When a small rotational perturbation (Ω) is applied to a system biased at a second-order EP, the resulting eigenfrequency splitting becomes proportional to the square root of the perturbation (4):

$$\Delta\omega \propto \sqrt{\Omega} \tag{4}$$

For ultra-low rotation rates ($\Omega \ll 1$), the square-root response provides a giant scale-factor enhancement ($\sqrt{\Omega} \gg$). This transition from a linear to a fractional-power response allows anti-PT symmetric gyroscopes to achieve orders-of-magnitude higher sensitivity to micro-rotations compared to strictly linear conventional architectures, making them a highly promising frontier for next-generation, miniaturized ultra-sensitive rotation sensors.

Transceivers [190,191], passive splitters [192], and SiOB technologies [185] simplify assembly. Simplified two-axis schemes [193,194] and differential architectures with two sources [195] reduce the cost. New architectures, such as cascade resonators (CROW), improve thermal stability [196] and sensitivity [188]. Compact technologies such as optomechanical gyroscopes (OMHSRG) and semiconductor ring lasers (SRLG) effectively circumvent the challenges associated with optical coupling and frequency lock-in [42,104,197–200].

6.4. Applied Aspects and Algorithmic Processing

In seismology, "giant" FOGs and rotary seismometers achieve a sensitivity of 10^{-9} rad/s/$\sqrt{\text{Hz}}$ [69,201,202]. Radiation-resistant fibers (PM-PCF) and fault-tolerant quad configurations have been developed for space missions [65,67,133,138,203]. In MWD and microtunneling, hybrid algorithms (ZUPT, CEEMDAN-LWT) provide accurate navigation in vibration conditions [63,96,204–207]. Optical accelerometers and FBG sensors demonstrate high reliability and accuracy in geotechnics [208,209], as well as in tactile sensors [210] and sensors based on LMR [211].

6.5. Signal Feedback (DSP and AI)

The usage of FPGA technology makes it possible to achieve a nanosecond system response and implement deep noise filtering [212–216]. Neural network algorithms are being successfully implemented for the tasks of technical diagnostics [217] and speeding up the procedure of the initial exhibition [218]. The accuracy of error prediction can be improved by using unified stochastic models [219]. Advanced modulation methods (random, multiharmonic, bipolar, DEHoI) linearize the output signal and extend the dynamic range [135,152,220–224]. A set of methods, including the adaptation of feedback coefficients [225], the use of the Smith predictor [226], the suppression of acousto-optical resonances [227], as well as parametric identification [228] and the use of triple circuits [229], ensures high robustness of the device. An additional increase in dynamic range is provided by hybrid control schemes [230].

6.6. Calibration and New Principles

Methods of automatic measurement of natural frequency (SHD) [231,232] and dynamic search of the north have been developed [233–235]. Atomic interferometers [236] and quantum gyroscopes [237] demonstrate the potential to overcome classical limits of accuracy. Brillouin gyroscopes (BLG/B-FOG) solve problems of nonlinearity through resonance stabilization [238–242]. To increase the linearity of the scale factor, methods of symmetric reset [243] and phase wavelength compensation [244] are used. New testing methods include the use of the Earth's rotation [245], autocollimators [235], and stochastic resonance for weak signals [246]. Intensity switching methods have been developed to reduce ARW [247]. Optimization of the phase shift also gives an ARW gain of up to 10 dB [248]. WGM microresonators demonstrate high quality [73,249,250], and low errors when optimizing bias [251] and suppressing non-linearities [252]. Optical effects in rotating WGM microresonators are also being studied [253]. New architectures have been proposed: anti-PT circuits [190] and dual-circuit structures (TLS) [254,255].

6.7. Interconnection Challenges and Splice Loss Reduction

While Hollow-Core Photonic Crystal Fibers (HC-PCFs) offer groundbreaking advantages for FOGs–such as drastically reduced Rayleigh scattering, Kerr effect, and thermal sensitivity–their integration into practical systems faces significant engineering hurdles. The primary obstacle is the interconnection between the HC-PCF coil and the conventional $LiNbO_3$ MIOC. Standard MIOCs are pigtailed with solid-core Single-Mode Fibers (SMFs). Splicing a solid-core SMF directly to a micro-structured HC-PCF results in prohibitive insertion loss and severe multi-path interference due to Mode Field Diameter (MFD) mismatch and the disruptive glass-to-air dielectric interface. To transition HC-PCF FOGs from laboratory prototypes to industrial applications, recent research has focused on minimizing the total system insertion loss. Current literature highlights some primary experimental strategies: mode field adapters, advanced cold splicing, micro-optic collimator coupling. Mode field adapters technique involves splicing a transitionary fiber segment–often a Graded-Index fiber or a thermally expanded core fiber–between the SMF and the HC-PCF to gradually match the MFDs. Recent state-of-the-art demonstrations have reduced the total SMF-to-HC-PCF-to-SMF chain insertion loss to an impressive range of 0.5 dB to 1.2 dB [256,257]. To entirely avoid the physical complexities of fusing solid glass to an air-core structure, some industrial designs utilize free-space coupling. Miniature C-lenses with anti-reflective coatings are used to collimate the beam between the SMF pigtail and the HC-PCF. While this introduces alignment challenges, it eliminates the direct glass–air fusion interface, achieving typical insertion losses of 0.5 dB to 2 dB. Crucially, this method provides superior suppression of back-reflection (often better than −50 dB), which is vital for maintaining the bias stability of the FOG [258–260]. These recent advancements indicate that while interconnection remains a complex engineering challenge, the insertion losses are rapidly approaching levels acceptable for navigation-grade FOG mass production.

7. Discussion

An analysis of the publications suggests that FOG technology has reached a turning point. Traditional IFOGs have almost exhausted the potential for physical sensitivity improvement. They provide navigational accuracy, but their weight and size characteristics and cost become a barrier to further distribution. On the other hand, the active development of integrated photonics and digital signal processing offers new ways to optimize the parameters of SWaP-C (dimensions, weight, power consumption, price). The main technological contradictions and trends identified during the research are discussed below.

7.1. From Hardware Compensation to Software Correction

For a long time, FOG accuracy has been improved due to its technical design. Engineers complicated the winding of coils (quadrupole/octupole) to minimize Shupe effect, using expensive polarized fibers (PM) and precision light sources [84,141,255]. However, modern research [7,23,25,102,110–116] records a paradigm shift in favor of algorithmic solutions. The implementation of deep learning methods (LSTM and RNN architectures) makes it possible to model and compensate for complex nonlinear temperature hysteresis. High accuracy of the device can now be obtained on the basis of standard, cheaper fibers. In such cases, error correction is performed by the microprocessor. Complex neural networks require computing power, which runs counter to the energy-saving requirements of autonomous systems (nanosatellites or IoT). Given the above, the vector of upcoming developments will shift towards optimizing neural network models for their operation on FPGAs in real time with minimal energy consumption.

7.2. Hollow Fibers as a Response to Shupe Effect

Despite the progress in thermal stabilization, Shupe effect remains a key factor limiting the long-term stability of the device. The complete solution to the problem lies not in dealing with the consequences, but in changing the physical medium of signal transmission. Using photonic crystal fibers with a hollow core (Air-Core/Hollow-Core PCF) [20,21,73] looks like the most promising direction. The propagation of light in an air or vacuum environment reduces the thermo-optical coefficient of the system by several orders of magnitude. This makes the gyroscope practically indifferent to temperature fluctuations and ionizing radiation. The main obstacle to mass adoption remains technological complexity: the coupling of hollow fibers with classical optics is associated with high signal losses. The creation of reliable methods for integrating such fibers with $LiNbO_3$ chips will be a crucial step towards the emergence of a new generation of strategic IFOG.

7.3. Integration and Miniaturization: IFOG vs. RFOG

Micro-IFOGs based on multifunctional integrated chips (MIOCs) [136,177] have already established themselves as a mature technology. They provide tactical accuracy and high reliability, successfully displacing MEMS sensors in harsh operating conditions. RFOGs based on silicon or silicon nitride resonators [180,185] theoretically benefit in size, allowing the creation of a chip-scale device. However, in practice (see Section 5.3) they suffer from coherent backscatter noise. Suppressing these noises requires cumbersome strapping and complex laser stabilization, which often negates the gain in the size of the most sensitive element. At the current stage, hybrid integration ($LiNbO_3$ + InP + Silica) looks like the optimal compromise. Fully monolithic silicon solutions still require refinement of technological processes to reduce losses in waveguides.

7.4. Positioning in the Sensor Ecosystem

The rigid distinctions between gyroscope categories outlined in Table 6 are increasingly converging. While MEMS technologies continue to improve in precision, and emerging optical modalities such as those based on the Brillouin effect and atomic interferometry [236,240] encroach upon the high-accuracy domain traditionally held by Ring Laser (RLG) and Hemispherical Resonator Gyroscopes (HRG), FOG technology retains critical design superiorities. Specifically, the absence of mechanical dither mechanisms (unlike RLGs) and the geometric flexibility of the fiber coil allow for conformal integration into complex, irregular volumes. This capability is decisive for high-density, space-constrained platforms. Ultimately, the future trajectory of FOG development will be defined by technological synergy: the fusion of novel materials (PCF), the miniaturization offered by hybrid photonic integrated circuits, and the deployment of deep learning algorithms for residual error compensation.

Table 6. Comparative analysis of the main sources of noise and errors in the FOG.

Noise Source	Physical Mechanism	Impact on Performance	Hardware Solutions	Advanced/AI Solutions
Shupe Effect	Changes in the refractive index (n) and fiber length (L) due to temperature gradients disrupt the reciprocity of the paths of the oncoming rays	Causes a zero-offset drift (bias drift), which is mistakenly interpreted as rotation	Special types of coil winding (quadrupole, octupole). The use of air-core fibers	Application of neural networks (LSTM, RNN) for hysteresis modeling. Active thermal stabilization on a chip
Polarization Noise	Cross-coupling of modes: energy transfer to orthogonal polarization due to fiber defects or external stresses	It leads to bias instability and drift due to interference of parasitic modes with the main signal	The use of PM fibers (Panda, Bow-tie), polarizers with a high extinction coefficient and welding with axis rotation by 90°	Polarimetric schemes (P-FOG) with analysis of Stokes parameters. Active self-compensation by modulators
Rayleigh Backscattering	Light scattering on microscopic inhomogeneities of glass density inside the fiber	Generates coherent noise that increases the ARW and worsens the sensitivity threshold	Using broadband light sources (SLD) with low coherence length to break interference	Phase modulation and "period superposition" methods. The use of composite structures and digital noise filtering

8. Conclusions

The evolution of FOG technology over the last ten years marks a transition from experimental prototypes to the dominant solution for high-precision navigation. These sensors are now indispensable for operations in signal-deprived zones. Historic barriers, including temperature-induced drift and Rayleigh backscattering, are being overcome by a shift in component design most notably through the use of Hollow-Core fibers (HC-PCF) and broadband sources that decouple the optical path from external perturbations.

Simultaneously, software has become as vital as optics. Advanced neural networks (RNN, LSTM) are taking over the calibration of nonlinear errors, rendering navigation-grade performance more cost-effective. Given their solid-state reliability and vibration immunity, FOGs remain the default choice for critical applications. Looking forward, the trajectory of inertial navigation will be defined by the intersection of integrated photonics, novel materials, and AI. These multidisciplinary fusion promises to yield a new class of compact, affordable devices capable of resilient operation under any conditions.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial Intelligence
Anti-PT symmetry	Anti-Parity Time symmetry
ARW	Angle Random Walk
B-FOG	Brillouin Fiber-Optic Gyroscope
BIC	Bound States in the Continuum
BLG	Brillouin Laser Gyroscope
CCW	Counter-Clockwise
CEEMDAN-LWT	Complete Ensemble Empirical Mode Decomposition with Adaptive Noise Lifting Wavelet Transform
CFD	Computational Fluid Dynamics
CMA-ES	Covariance Matrix Adaptation Evolution Strategy
CROW	Coupled Resonator Optical Waveguides
CW	Clockwise
DEHoI	Differential Error Hamiltonian-oriented Integration
DSP	Digital Signal Processing
EP	Exceptional Point
FEM	Finite Element Method
FOG	Fiber-Optic Gyroscope
FPGA	Field-Programmable Gate Array
GNSS	Global Navigation Satellite Systems
GPS	Global Positioning System
HRG	Hemispherical Resonator Gyroscope

HC-PCF	Hollow-Core Photonic Crystal Fibers
IFOG	Interferometric Fiber-Optic Gyroscope
InP	Indium Phosphide
INS	Inertial Navigation System
IOC	Integrated Optical Circuit
IoT	Internet of Things
LiNbO3	Lithium Niobate
LMR	Lossy Mode Resonance
LSTM	Long Short-Term Memory
MCF	Multi-Core Fiber
MEMS	Micro-Electro-Mechanical Systems
MFD	Mode Field Diameter
Micro-IFOG	Micro Interferometric Fiber-Optic Gyroscope
MIOC	Multifunctional Integrated Optical Circuit
MPE-CEEMDAN	Multiscale Permutation Entropy Complete Ensemble Empirical Mode Decomposition with Adaptive Noise
MWD	Measurement While Drilling
OMHSRG	Opto-Mechanical Hemispherical Solid-State Resonator Gyroscope
PCF	Photonic Crystal Fiber
PF	Polynomial Fitting
PDH-modulation	Pound-Rever-Hall modulation
P-FOG	Polarization-maintaining Fiber-Optic Gyroscope
PM	Polarization-Maintaining
PMF	Polarization-Maintaining Fibers
PM-PCF	Polarization-Maintaining Photonic Crystal Fiber
PSK	Phase Shift Keying
PSO	Particle Swarm Optimization
RFOG	Resonant Fiber-Optic Gyroscope
RIN	Relative Intensity Noise
RLG	Ring Laser Gyroscope
RMS	Root Mean Square
RMSE	Root Mean Square Error
RNN	Recurrent Neural Network
RPCFOG	Resonant Photonic Crystal Fiber-Optic Gyroscope
SHD	Step-Height Detection
SiN	Silicon Nitride
SiOB	Silicon Optical Bench
SLD	Superluminescent Diode
SMF	Single-Mode Fiber
SPR	Surface Plasmon Resonance
SRLG	Semi-Rigid Laser Gyroscope
SVR	Support Vector Regression
SWaP-C	Size, Weight, Power, and Cost
TLS	Two-Level Systems
ToF	Time of Flight
UAV	Unmanned Aerial Vehicle
VMD	Variational Mode Decomposition
WGM	Whispering Gallery Mode
ZUPT	Zero Velocity Update

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